Cheap Multi-Robot Autonomous Fleets for Disaster Recovery: Phone-as-Brain Rovers with Depth Sensing, Obstacle Avoidance, and Fleet Command Interfaces

Yojan Gautam

October 24, 2026

Contents

Introduction **2**

1 Background and Related Work **7**

1.1 Robots in disaster response 7

1.1.1 World Trade Center, 2001 7

1.1.2 Fukushima Daiichi, 2011 7

1.1.3 DARPA Subterranean Challenge, 2021 8

1.1.4 Champlain Towers South, Surfside, 2021 8

1.1.5 Recurring lessons 8

1.2 Levels of autonomy 8

1.3 Supervising multiple robots 9

1.3.1 Neglect tolerance and fan-out 9

1.3.2 Attention, workload, and situation awareness 10

1.4 Operator interfaces for robot fleets 10

1.5 Communication constraints 10

1.6 Low-cost onboard navigation 11

1.6.1 Local planning 11

1.6.2 Phones as robot computers 11

1.7 Multi-robot coordination 11

1.8 Gaps addressed by this thesis 11

Introduction

Robots in real missions

Robots have been part of disaster response for more than two decades, and the same problems keep returning. At the World Trade Center in 2001, every robot on the pile was teleoperated. The one robot that was lost and not recovered was a wireless unit: it lost its control link and had no onboard intelligence to move and reacquire the signal [17]. At Fukushima Daiichi in 2011, the concrete walls of the reactor buildings were expected to block radio, so the Quince robot was driven over a cable, and teleoperation had to be simple enough for first responders rather than the researchers who built it [20]. At the DARPA Subterranean Challenge Final in 2021, each team was allowed a single Human Supervisor to interact with its robots, and one operator managed eleven robots in the Louisville Mega Cavern [7, 1]. At the Champlain Towers South collapse in Surfside, Florida, the same year, drones flew 304 missions for mapping and strategic operations alone, and the drone teams coordinated through a WhatsApp group chat [19].

Read together, these deployments show three recurring failures. Communication fails, and a robot that depends entirely on its link stops being useful the moment the link drops. One person ends up juggling several robots with no shared picture of where they are and what they need. The robots carry too little autonomy for the job, or an amount chosen by trial and error rather than by measurement. SubT showed that one-person supervision of a fleet is possible. It also showed its limits: the operator who ran eleven robots described them as "quite a lot to be managed by a single human operator under a constrained time limit," and operators across teams were still the bottleneck for confirming finds, retasking robots, and freeing stuck ones [1].

One operator, many robots

This thesis studies the setting those missions point toward: one human operator supervising a fleet of ground robots. Three questions make that setting hard.

1. **Autonomy.** What is the minimum level of autonomy each robot needs to complete a mission, and when should the human step in?

2. **Attention.** When several vehicles need the operator at the same time, how should the operator's attention be scheduled across them?

3. **Bandwidth.** How much telemetry can each robot send as the fleet grows, and what should the operator see continuously versus on demand?

These questions interact. More onboard autonomy means each robot asks for help less often and can survive a lost link, which frees operator attention and reduces the bandwidth each robot needs. More autonomy also costs development effort, compute on the vehicle, and operator trust, and a robot that is more autonomous than the mission needs is harder to understand when it misbehaves. The practical goal is the lowest level of autonomy that finishes the job and survives a lost link, with a human takeover always one action away.

Levels of autonomy

To make the autonomy question measurable, this thesis uses a six-level ladder for a single ground vehicle (Table 1). Each level adds one responsibility to the robot and removes one from the human.

Table 1: Autonomy levels used in this thesis. Runtime names are the values the ARGOS operator application publishes to Terra.

Level	Name	Robot does	Runtime name
L0	Full teleoperation	Nothing on its own; the human controls all motion.	`teleop`
L1	Assisted teleoperation	The human drives; the rover prevents collisions.	`assisted_teleop`
L2	Waypoint, no avoidance	Drives from A to B along predefined waypoints.	`waypoint_direct`
L3	Waypoint, with avoidance	Navigates from A to B and stops for or avoids obstacles.	`waypoint`
L4	Search	Searches an area autonomously and detects targets.	`target_search`
L5	Decision making	Decides what is needed now and what is best for the fleet.	**[TODO: not defined]**

The ladder is a working taxonomy for this platform, not a general standard. Chapter 1 places it among existing level-of-autonomy frameworks. An earlier version of this project used a different ladder, with "stop and ask" as its own level; runtime names and research labels are not interchangeable, so every study condition records the exact software revision and behavior used. **[TODO: Decide whether `supervised` autonomy, which the operator application also offers, is a separate level or a mode across L3–L4.]**

The phone as the robot's brain

A fleet is only practical if each vehicle is cheap enough to field, lose, and replace, and simple enough to power, update, and send out again under time pressure. Robots that navigate in clutter usually carry a dedicated onboard computer cabled to a separate sensor suite. That computer makes perception and planning straightforward to host, but it is also a logistics item: bill-of-materials cost, a thermal and power design, spare parts, and a machine image to flash and maintain on every unit.

A current consumer phone already holds what a small rover needs for local navigation: cameras, an inertial measurement unit, radios, a battery, and a processor able to run real-time vision, often with a depth-capable sensor. The premise of this thesis is to make that phone the vehicle's onboard brain. An operator installs an application, mounts the phone on a low-cost differential-drive chassis, and the phone senses, estimates its pose, plans, and commands the wheels. In the current prototype, a Raspberry Pi receives wheel commands from the phone over Bluetooth and drives the motors; the long-term plan replaces it with a low-cost microcontroller attached to the phone over USB.

Research questions

The central question of the thesis is:

> **RQ.** What minimum level of local autonomy allows one human operator to supervise an increasing number of robots effectively?

Three sub-questions, each the subject of one paper, support it.

RQ1 (phone as brain). Can a consumer phone run the sensing and control loop for a low-cost differential-drive rover well enough for waypoint navigation with local obstacle avoidance, without onboard industrial compute?

RQ2 (shared maps). How well can several phone-mounted rovers build and fuse a shared map under limited bandwidth and noisy depth, compared with single-robot mapping?

RQ3 (supervision). What operator interface and attention scheduling let one person supervise N rovers as onboard autonomy increases, and which vehicle should receive human attention first?

The primary experimental factors are fleet size N and autonomy level A.

Approach

The work is built as three systems that share one robotics core.

- **Terra** is the vehicle. A shared Rust core holds state estimation, local occupancy mapping, planning, the control-authority arbiter, and velocity control. The TerraPhone iOS application wraps that core, reads the camera, pose, depth, and IMU through ARKit and Core Motion, and sends wheel effort to the motor controller. Terra keeps motion authority on the vehicle: it checks command freshness and arbitrates between teleoperation, assisted teleoperation, waypoint autonomy, and supervised autonomy.
- **ARGOS** (Adaptive Robotic Group Operator System) is the operator layer. It is a native desktop application that shows the fleet on one map with each rover's GPS fix and accuracy, sends autonomy levels, goals, and held teleoperation, and keeps stop and L0 manual control one click away. ARGOS sends intent; it does not stream low-level velocity as its normal command.

- **Zorvane** is the simulator. It runs the same Terra algorithms, so a mission can be supervised in simulation and on real rovers through the same ARGOS interface.

ARGOS communicates with real rovers and with Zorvane over Zenoh, a publish–subscribe protocol [**TODO: cite Zenoh**].

The planned study has operators supervise rovers through ARGOS in real-world missions and in Zorvane, on tasks such as finding a target object, scouting a route, or covering a sector before a deadline. Two conditions frame it. In the *fixed* condition, one autonomy level is locked for the whole run, which gives a baseline per level. In the *free* condition, the operator may switch levels at any time, and the system logs when, why, and in what context; the operator's choices are the core result. Conditions vary communication loss, the number of robots per operator, debris, deadlines, and unplanned events. Measures are mission success, completion time, operator workload, interventions, and time spent at each level. The design is an early proposal and will be made more rigorous before any human-subject study [**TODO: ethics / IRB approval, sample size, counterbalancing, workload instrument**].

Current status

This section states what exists so that later chapters can separate implemented work from planned work. As of October 2026:

- L0–L3 run in simulation, and L3 has been demonstrated on the real rover. L3 obstacle avoidance runs on the phone from live depth, with a local occupancy map and a voxel map, and has been demonstrated indoors against cables, chair legs, carts, and hallway walls [**TODO: report as a qualitative demonstration until measured trials exist**].
- The ARGOS desktop application provides a fleet map, autonomy selection, waypoint goals, teleoperation, emergency stop, and a first-person video panel, and is connected to Zorvane.
- L4 target search exists in Terra and has passed a deterministic simulator replay; physical detection and hardware validation are outstanding.
- The operator pilot study has not been run. No human-subject results exist yet.

Contributions

The planned contributions are:

1. A phone-as-brain ground rover in which a consumer phone is the only onboard computer for sensing, pose estimation, mapping, planning, and control, with a shared Rust core that runs unchanged in simulation.
2. An autonomy ladder for a supervised ground fleet, implemented as selectable runtime levels with a single onboard arbiter for control authority and takeover.

3. ARGOS, a one-operator fleet supervision interface with a shared fleet map, one-click autonomy switching, and one-click takeover.

4. An experimental study of how much autonomy missions need and when operators choose to intervene, as fleet size, communication quality, and environment vary. [**TODO: Fill in once the study is run; do not claim findings before then.**]

Scope

The thesis treats a homogeneous fleet of Terra ground rovers supervised by one operator. Heterogeneous fleets, aerial vehicles, and multiple cooperating operators are outside its scope. The main experimental scope is L0–L3; L4 is evaluated where the platform supports it, and L5 is discussed as future work. Disaster response is the motivating application. The same cost and deployability constraints apply to fire and rescue and to surveillance, but this thesis does not evaluate those settings separately.

Thesis outline

[**TODO: Confirm chapter list and numbering against the final structure.**] Chapter 1 reviews robots in disaster response, levels of autonomy, supervisory control of multiple robots, operator interfaces, communication constraints, and low-cost navigation, and identifies the gaps this thesis addresses. Chapter 2 describes the platform: the Terra rover, the phone-as-brain software stack, and the Zorvane simulator. Chapter 3 describes onboard autonomy from obstacle avoidance through search. Chapter 4 presents the ARGOS operator interface. Chapter 5 reports the evaluation. Chapter 6 concludes and outlines future work.

Chapter 1

Background and Related Work

This chapter places the thesis in five bodies of work: robots in real disaster response, levels of autonomy, supervisory control of multiple robots, operator interfaces for fleets, and low-cost onboard navigation. It closes with the gaps that motivate the platform and study in later chapters.

1.1 Robots in disaster response

Rescue robotics has a field record that is unusual in robotics: robots have been deployed at real incidents, and the deployments have been documented. Murphy's account of the field gives the broad picture [18]; this section concentrates on four missions whose lessons recur throughout the thesis.

1.1.1 World Trade Center, 2001

The response at the World Trade Center was the first major use of robots for urban search and rescue. All of the robots used were teleoperated. The only robot lost and not recovered was wireless: it lost its control link, and with no onboard intelligence it could not move to reacquire the signal [17]. The lesson for this thesis is that autonomy has to live on the robot as well as in the link. A robot whose only behavior is to wait for commands fails as soon as the commands stop.

1.1.2 Fukushima Daiichi, 2011

After the Fukushima Daiichi accident, the Quince robot was sent into the reactor buildings. Thick concrete walls were expected to block radio, so Quince used a cable link (VDSL) instead of wireless communication [20]. The robot's developers could not operate it on site themselves, so its teleoperation had to be easy for first responders to learn [20]. **[TODO: Add page or section numbers for both claims.]** Two lessons follow. Communication cannot be assumed, and in some environments the choice is between a tether and a robot that can act alone. The operator is often not the expert who built the robot, so the interface has to be simple.

1.1.3 DARPA Subterranean Challenge, 2021

The DARPA Subterranean Challenge (SubT) asked teams to explore, map, and search underground tunnels, urban structures, and caves with robot teams. Its final event took place in September 2021 in the Louisville Mega Cavern [7]. SubT is the closest existing evidence for one-operator fleet supervision under field conditions: each team had a single Human Supervisor, who was the only person allowed to interact with the robots during a run [1]. One operator, Chao Cao, managed eleven robots [1].

Operator interviews after the final show both that this worked and where it strained [1]. Operators were still the bottleneck: they confirmed artifact reports, retasked robots, and got stuck robots free. Team CoSTAR reported that low-level commands were "too difficult for the operator," so over the years of the program the team moved them onto the robots. Operator Marco Tranzatto asked for "a more centralized way to give commands to the robots." Operator Martin Dlouhý observed that "it's obvious to a human when the robot is not doing well," which argues for keeping a fast human takeover. IEEE Spectrum also noted that relying on a skilled, practiced operator suits a competition team but could be impractical for robots deployed operationally [1].

SubT therefore leaves open the question this thesis studies. Teams arrived at their autonomy levels through years of iteration, not through a measured account of how much autonomy a mission needs and when the human should step in. **[TODO: Add SubT team system papers (for example CERBERUS, CoSTAR, CSIRO Data61) for their autonomy and communication architectures.]**

1.1.4 Champlain Towers South, Surfside, 2021

After the Champlain Towers South collapse in June 2021, drones flew 304 missions for strategic operations alone [19]. The drone teams coordinated through a WhatsApp group chat, and mobile apps on the rubble pile showed GPS errors as large as ten meters [19]. The robots themselves worked; what was missing was a shared operational picture of where each vehicle was, how well it was localized, and who was tasking it.

1.1.5 Recurring lessons

Table 1.1 summarizes the failures in these missions and the design response this thesis takes to each. The right-hand column states a design intent, not a claim that ARGOS would have solved these missions.

1.2 Levels of autonomy

Describing how work is split between a human and an automated system by discrete levels has a long history. Sheridan and Verplank proposed a ten-level scale running from a human doing everything to a computer acting entirely on its own [27]. Parasuraman, Sheridan, and Wickens refined this by noting that automation can be applied at different levels to each of four stages: information acquisition, information analysis, decision selection, and action implementation [23].

Table 1.1: Failures in documented robot deployments and the corresponding design intent in this thesis.

Mission	What went wrong	Design intent in this thesis
WTC 2001	Wireless robot lost its link; no onboard autonomy to recover.	Put autonomy on the robot (L1–L3), not only on the link.
Fukushima 2011	Concrete blocked radio; teleoperation had to be easy for responders.	One simple operator application, with L0 manual through L3 waypoint one click apart.
SubT 2021	One supervisor ran eleven robots but remained the bottleneck; autonomy set by trial and error.	Measure how much autonomy a mission needs and when the human should step in; switch levels in one click.
Surfside 2021	304 missions coordinated by group chat; phone GPS errors up to 10 m.	One shared fleet map with each robot, its GPS fix, and its reported accuracy.

Beer, Fisk, and Rogers adapted these ideas to robotics and to the sense–plan–act structure of a robot, and argued that the appropriate level depends on the task and the environment [2]. For road vehicles, SAE J3016 defines six levels of driving automation [25].

The ladder in Table 1 differs from these in two ways. First, it is defined by mission behaviors on one platform (teleoperate, assist, follow waypoints, avoid, search, decide), so that each level is a selectable runtime mode that can be held fixed for an experimental condition. Second, its upper levels move from a single vehicle's behavior to fleet-level decisions, which general automation scales do not address. **[TODO: Compare with ALFUS (Autonomy Levels for Unmanned Systems, NIST) and cite it once the source is confirmed.]**

A fixed level is also not the only option. Work on adjustable, sliding, or mixed-initiative autonomy lets the level change during a mission, by the human or by the system [12]. **[TODO: Add a survey of adjustable / sliding autonomy and mixed-initiative control.]** The free-switching condition of the planned study belongs to this line of work: it treats the operator's choice of level as data.

1.3 Supervising multiple robots

1.3.1 Neglect tolerance and fan-out

Supervisory control describes a human who sets goals and monitors an automated system rather than controlling it continuously [26]. For robot teams, Olsen and colleagues gave this a quantitative form. *Neglect time* is how long a robot can be left alone before its performance drops below an acceptable level, and *interaction time* is how long the human needs to service it [21]. *Fan-out* is the number of robots one person can effectively operate at once, estimated from those two quantities [22]. Crandall and colleagues tested such interaction schemes in multitasking environments [5], and Crandall and Cummings examined which metrics predict supervisory performance with multiple robots [4]. Cummings and Mitchell modeled operator capacity for supervising multiple unmanned aerial vehicles and included the time robots spend waiting for the operator [6].

This literature frames the thesis question directly. Raising the autonomy level should lengthen neglect time and shorten interaction time, which raises fan-out. What it does not

settle is how that relationship looks on a real low-cost platform under communication loss and clutter, and whether operators, when free to choose, pick the level that the model would predict.

1.3.2 Attention, workload, and situation awareness

When several robots need help at once, the operator must decide whom to serve first. Each robot that waits accumulates wait time, and the operator must also switch context between robots. Chen and Barnes review human factors issues in multirobot control, including operator workload, situation awareness, trust, and the effect of automation on both [3]. Situation awareness, in Endsley's definition, covers perceiving the elements of the environment, understanding them, and projecting their future state [10]. Workload is commonly measured with the NASA Task Load Index (NASA-TLX) [14], the instrument currently proposed for this thesis. **[TODO: Add work on attention allocation and scheduling of operator attention across robots (for example queuing models of operator service).]** **[TODO: Add work on interruptions and task switching cost in multi-robot supervision.]**

1.4 Operator interfaces for robot fleets

The deployments above show what operators ask for: one centralized map from which to command robots [1], and a single shared picture in place of chat-based coordination [19]. Interfaces for multi-robot supervision commonly combine a map-centric overview with on-demand detail for a selected robot. Mixed-reality and immersive command interfaces extend this; Milgram and Kishino's reality–virtuality continuum is the standard way to classify them [15]. **[TODO: Replace the placeholder survey of AR/VR command interfaces for multi-robot supervision (`chen2018survey` in thesis_outline.tex) with a confirmed source.]** **[TODO: Add examples of fleet supervision interfaces from SubT teams and from multi-UAV ground control stations.]**

1.5 Communication constraints

Every deployment in Section 1.1 met a communication limit: a lost wireless link at WTC, radio blocked by concrete at Fukushima, and underground radio limits throughout SubT. Two design responses follow. The first is onboard autonomy that keeps the robot safe and useful when the link drops. The second is a telemetry design that sends low-bandwidth state continuously (pose, mission state, health, autonomy level, and whether the robot needs help) and sends high-bandwidth data such as video, point clouds, and maps only when the operator asks for it. **[TODO: Cite communication-aware autonomy and multi-robot networking work, including SubT communication architectures (mesh radios, deployable relay nodes).]** **[TODO: Cite work on bandwidth scaling of telemetry with fleet size.]**

1.6 Low-cost onboard navigation

1.6.1 Local planning

Local obstacle avoidance for differential-drive robots is well established. The dynamic window approach searches the velocities a robot can reach within one control cycle, discards those that lead to collision, and scores the rest on heading toward the goal, clearance, and forward speed [11]. Because it replans every cycle, a sequence of obstacles updates the trajectory without a single precomputed path. [**TODO: Add other local planners used for comparison (for example timed elastic bands, vector field histogram), if the methods chapter uses them.**]

1.6.2 Phones as robot computers

Phones have been used in robotics as teleoperation clients, communication links, and extra cameras. [**TODO: Cite prior smartphone-based robots and phone-as-controller platforms.**] Using the phone as the vehicle's only onboard computer depends on how well it estimates its own motion. Visual–inertial odometry (VIO) fuses camera and IMU data for pose; Delmerico and Scaramuzza benchmarked monocular VIO algorithms on flying-robot hardware [8]. [**TODO: Add the phone VIO references from the literature review (Kim 2022; Marino 2022) with full details.**] On iOS, ARKit provides camera pose and, on supported devices, depth. On Android, ARCore provides motion tracking that combines visual feature points with the IMU, and depth maps from the main RGB camera on supported devices [13]. A single Rust library can serve both platforms through UniFFI, which generates Swift and Kotlin bindings [16].

1.7 Multi-robot coordination

At the upper end of the autonomy ladder, robots coordinate among themselves. ALLIANCE is a behavior-based architecture for fault-tolerant cooperation, in which robots take over tasks that teammates fail to complete [24]. Market-based approaches allocate tasks through auctions and bids among robots [9]. Collaborative mapping across several robots is the subject of Paper 2 of this thesis. [**TODO: Add collaborative / multi-robot SLAM references for Paper 2.**]

1.8 Gaps addressed by this thesis

The work reviewed above leaves four gaps that this thesis targets.

1. **Measured autonomy requirements.** Field teams, including at SubT, chose autonomy levels through iteration. Fan-out models predict the effect of autonomy on supervision, but there is little measurement of the minimum level real missions need, or of when operators choose to step in when they may switch freely. [**TODO: Confirm this claim against the literature before submission.**]

2. **Cost and deployability.** Platforms with dedicated onboard computers are expensive to field in numbers. A phone-as-brain rover tests whether commodity hardware can carry L0–L3 autonomy.

3. **An integrated system.** Onboard avoidance, an onboard authority arbiter, and the operator interface are usually studied separately. This thesis builds them as one system, so that a change of level in the interface maps directly to a change of behavior on the robot, in simulation and on hardware.

4. **A simple shared operator picture.** Field operators asked for one centralized map and simple controls. ARGOS provides a single fleet map with one-click autonomy switching and takeover, designed for operators who are not the robot's developers.

The next chapter describes the platform that supports this study.

Bibliography

[1] Evan Ackerman. DARPA SubT finals: Robot operator wisdom. IEEE Spectrum, 23 September 2021. `https://spectrum.ieee.org/darpa-subterranean-challenge-operator`, 2021.

[2] Jenay M. Beer, Arthur D. Fisk, and Wendy A. Rogers. Toward a framework for levels of robot autonomy in human-robot interaction. *Journal of Human-Robot Interaction*, 3(2): 74–99, 2014. VERIFY.

[3] Jessie Y. C. Chen and Michael J. Barnes. Human-agent teaming for multirobot control: A review of human factors issues. *IEEE Transactions on Human-Machine Systems*, 44(1): 13–29, 2014. VERIFY.

[4] Jacob W. Crandall and M. L. Cummings. Identifying predictive metrics for supervisory control of multiple robots. *IEEE Transactions on Robotics*, 23(5):942–951, 2007. VERIFY.

[5] Jacob W. Crandall, Michael A. Goodrich, Dan R. Olsen, and Curtis W. Nielsen. Validating human-robot interaction schemes in multitasking environments. *IEEE Transactions on Systems, Man, and Cybernetics – Part A: Systems and Humans*, 35(4):438–449, 2005. VERIFY.

[6] M. L. Cummings and Paul J. Mitchell. Predicting controller capacity in supervisory control of multiple UAVs. *IEEE Transactions on Systems, Man, and Cybernetics – Part A: Systems and Humans*, 38(2):451–460, 2008. VERIFY.

[7] Defense Advanced Research Projects Agency. DARPA subterranean challenge winners. `https://www.darpa.mil/news/2021/subterranean-challenge-winners`, 2021. News release, September 2021.

[8] Jeffrey Delmerico and Davide Scaramuzza. A benchmark comparison of monocular visual-inertial odometry algorithms for flying robots. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 2502–2509, 2018. VERIFY pages.

[9] M. Bernardine Dias, Robert Zlot, Nidhi Kalra, and Anthony Stentz. Market-based multirobot coordination: A survey and analysis. *Proceedings of the IEEE*, 94(7):1257–1270, 2006.

[10] Mica R. Endsley. Toward a theory of situation awareness in dynamic systems. *Human Factors*, 37(1):32–64, 1995. VERIFY.

[11] Dieter Fox, Wolfram Burgard, and Sebastian Thrun. The dynamic window approach to collision avoidance. *IEEE Robotics & Automation Magazine*, 4(1):23–33, 1997.

[12] Michael A. Goodrich, Dan R. Olsen, Jacob W. Crandall, and Thomas J. Palmer. Experiments in adjustable autonomy. In *IJCAI Workshop on Autonomy, Delegation and Control: Interacting with Intelligent Agents*, 2001. VERIFY.

[13] Google. ARCore fundamental concepts. `https://developers.google.com/ar/develop/fundamentals`, 2026. Accessed October 2026.

[14] Sandra G. Hart and Lowell E. Staveland. Development of NASA-TLX (task load index): Results of empirical and theoretical research. In Peter A. Hancock and Najmedin Meshkati, editors, *Human Mental Workload*, pages 139–183. North-Holland, 1988. VERIFY.

[15] Paul Milgram and Fumio Kishino. A taxonomy of mixed reality visual displays. *IEICE Transactions on Information and Systems*, E77-D(12):1321–1329, 1994.

[16] Mozilla. UniFFI: A multi-language bindings generator for Rust. `https://mozilla.github.io/uniffi-rs/`, 2026. Accessed October 2026.

[17] Robin R. Murphy. Rescue robotics for homeland security. *Communications of the ACM*, 47(3):66–68, 2004. URL `https://cacm.acm.org/research/rescue-robotics-for-homeland-security/`. VERIFY volume, number and pages.

[18] Robin R. Murphy. *Disaster Robotics*. MIT Press, 2014. VERIFY.

[19] Robin R. Murphy et al. Robots and drones at the Surfside building collapse. IEEE Spectrum, August 2021. `https://spectrum.ieee.org/building-collapse-surfside-robots`, 2021. VERIFY exact title and author list.

[20] Keiji Nagatani et al. Redesign of rescue mobile robot Quince: Toward emergency response to the nuclear accident at Fukushima Daiichi nuclear power station on March 2011. In *IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, 2011. URL `https://k-nagatani.org/pdf/2011-SSRR-KEI-Rev-online.pdf`. VERIFY title, full author list and pages from the PDF.

[21] Dan R. Olsen and Michael A. Goodrich. Metrics for evaluating human-robot interactions. In *Performance Metrics for Intelligent Systems Workshop (PerMIS)*, 2003. VERIFY.

[22] Dan R. Olsen and Stephen Bart Wood. Fan-out: Measuring human control of multiple robots. In *ACM Conference on Human Factors in Computing Systems (CHI)*, pages 231–238, 2004. VERIFY.

[23] Raja Parasuraman, Thomas B. Sheridan, and Christopher D. Wickens. A model for types and levels of human interaction with automation. *IEEE Transactions on Systems, Man, and Cybernetics – Part A: Systems and Humans*, 30(3):286–297, 2000. VERIFY.

[24] Lynne E. Parker. ALLIANCE: An architecture for fault tolerant multirobot cooperation. *IEEE Transactions on Robotics and Automation*, 14(2):220–240, 1998.

[25] SAE International. Taxonomy and definitions for terms related to driving automation systems for on-road motor vehicles. Technical Report J3016, SAE International, 2021. VERIFY revision year.

[26] Thomas B. Sheridan. *Telerobotics, Automation, and Human Supervisory Control*. MIT Press, 1992. VERIFY.

[27] Thomas B. Sheridan and William L. Verplank. Human and computer control of undersea teleoperators. Technical report, MIT Man-Machine Systems Laboratory, 1978. VERIFY.